

SoC compatible 1T1C FeRAM memory array based on ferroelectric $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$

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ABSTRACT

This paper experimentally demonstrates fundamental memory array operation of a ferroelectric HfO_2 -based 1T1C FeRAM. Metal/ferroelectric/metal (MFM) capacitors consisting of a $\text{TiN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(\text{HZO})/\text{TiN}$ stack were optimized for a sub 500 °C process. Structures revealed excellent performance such as remanent polarization $2P_r > 40 \mu\text{C}/\text{cm}^2$, endurance $> 10^{11}$ cycles, and 10 years data retention at 85 °C. Furthermore, the MFM capacitors were successfully integrated into a 64 kbit 1T1C FeRAM array including our dedicated circuit for array operation. Back-end-of-line (BEOL) wiring showed no degradation of the underlying CMOS logic. Program and read operation were properly controlled resulting in 100 % bit functionality at an operation voltage of 2.5 V and operating speed at 14 ns. This technology matches requirements of last level cash (LLC) and embedded non-volatile-memory (NVM) in low power System-on-a-Chip (SoC) for IoT applications.

Keywords: hafnium oxide, zirconium oxide, capacitor

INTRODUCTION

Recently, ferroelectric HfO_2 -based FeRAM has been getting a lot of attention due to its high performance operation and CMOS compatible process [1-2]. Compared to ferroelectric FET (FeFET), 1T1C FeRAM has the advantage of higher endurance, lower voltage operation and disturb free operation [3]. However, a memory array operation using a dedicated circuit such as address decoders and a sense amplifier (SA) for the HfO_2 -based 1T1C FeRAM has never been reported. In this work for the first time, we fabricated a 64 kbit array using a ferroelectric HfO_2 -based 1T1C test chip with CMOS logic circuit and demonstrated good array operation yield. Furthermore, MFM capacitor size and operation condition dependency of the test chip on the read voltage margin supports the scalability of the SoC compatible LLC appreciations.

MFM CHARACTERISTICS

Intrinsic ferroelectricity of MFM capacitors was electrically confirmed on single large capacitors ($\varnothing 100 \mu\text{m}$). MFMs were fabricated including symmetrical TiN electrodes and 11 nm thick HZO deposited by ALD with 1:1 Hf:Zr ratio. Both 10 nm thick TiN electrodes were sputtered by PVD. The appearance of a ferroelectric orthorhombic phase was verified by grazing incidence X-ray diffraction (GIXRD) after a 500 °C crystallization anneal (Fig.1). A remanent polarization of $2P_r > 40 \mu\text{C}/\text{cm}^2$ was achieved as indicated in the PE plot (Fig.2). Fig. 3 shows the endurance performance up to 10^9 cycles at 100 kHz cycling frequency. Negligible small changes of polarization were obtained during the data retention test at 85 °C for 100 min (Fig.4). These results suggest excellent performance with high reliability for integrated 1T1C FeRAM.

1T1C ARRAY INTEGRATION

Fig.5 shows a cross section SEM image of the 64 kbit 1T1C FeRAM array. A MFM capacitor consisting of a $\text{TiN}/\text{HZO}/\text{TiN}$ stack was successfully integrated directly on top of the transistor allowing the crystallization anneal at 500°C prior to BEOL metal line processing. The capacitor process flow is shown in Fig. 6. A patterned bottom electrode (BE) is covered with HZO and the top electrode (TE). Fig.7 shows an optical microscope image of the 64 kbit test chip for ferroelectric HfO_2 -based 1T1C FeRAM. A dedicated circuit for 1T1C FeRAM operation with different size of capacitors ranging from 0.4 to $1.0 \mu\text{m}^2$ is integrated into the same chip.

64 KBIT 1T1C ARRAY DEMONSTRATION

Fig.8 shows the read operation scheme of the 1T1C cell. A voltage change of V_{BL} resulting from the transient capacitance difference (C_0 vs. C_1) of the two memory states of the ferroelectric capacitor (C_{FE}) in comparison to a reference capacitor (C_{REF}) is analyzed by an integrated SA using a step pulse sensing approach [5]. Simulation and experimental results of the detected V_{BL} shown in Fig.9 reveals that this device is ideally functioning to separate both memory states (data0 and data1). Fig.10 shows the distribution of V_{BL} obtained from both memory states for the 1T1C 64 kbit array based cells with a $1.0 \mu\text{m}^2$ capacitor size. A 100% bit functionality with a read voltage margin $> 0.48\text{V}$ at 4 sigma tail is detected demonstrating a large signal margin even for arrays having a significantly higher density. Fig. 11 indicates a shmoo plot for an operating time from 1 ns to 50 ns and an operation voltage from 2.5 V to 3.5 V for both read and write operations of the test chip. No fail bit is observed for a 14 ns write and a 8 ns sense operation at 2.5 V each. Fig.12 shows the dependency of the read voltage margin on the MFM capacitors size. Dashed lines were calculated from the formula described in Fig.8(b). The obtained read voltage margin was in good agreement with the estimation for a 130 nm technology. As a result, a read voltage margin $> 0.10\text{V}$ can be achieved using a $< 0.05 \mu\text{m}^2$ capacitor size which can be fabricated using a 40 nm technology based 3D cylinder capacitor[7].

CONCLUSION

We demonstrate a ferroelectric HfO_2 -based 1T1C FeRAM memory array suitable for system on chip (SoC) applications. MFM capacitors are successfully integrated into a 64 kbit array without any degradation of the underlying CMOS logic characteristics. For the first time, perfect yield and excellent operation performance for low voltages of 2.5 V at 14 ns operating speed was observed during array operation. This technology meets the performance of LLC and any other IoT applications of existing embedded memories as shown in table.1 in comparison to other technology options.

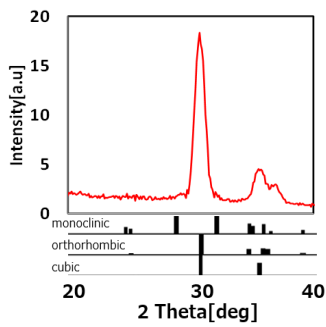


Fig. 1: GIXRD spectrum of TiN/HZO/TiN stack after 500 °C anneal

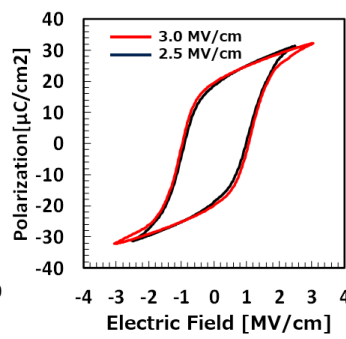


Fig. 2: Polarization vs field hysteresis of a MFM capacitor with applied fields of 2.5-3.0 MV/cm.

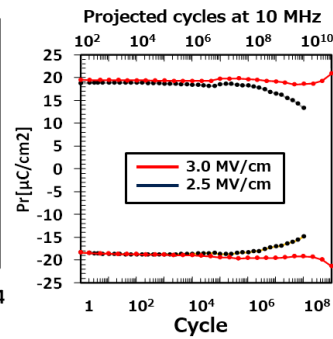


Fig. 3: Cycling endurance at 100 kHz (bottom scale) and projected endurance at 10 MHz (top scale) assuming same time to break down.

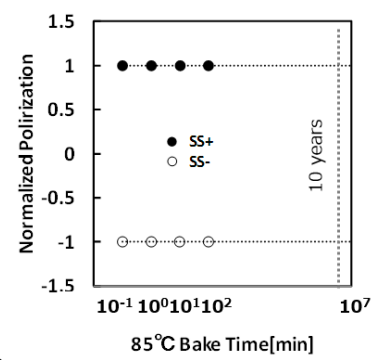


Fig. 4: Data retention up to 100 min baking time at 85 °C and projection to 10 years with Same State(SS) [4].

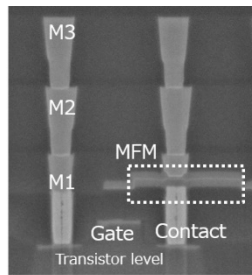


Fig. 5: Schematic cross section SEM image of MFM capacitors integrated within a BEOL process.

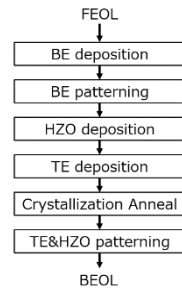


Fig. 6: Process flow to fabricate MFM below 500 °C BEOL budget.

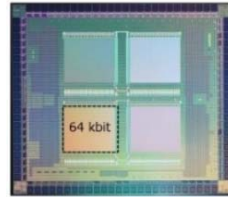


Fig. 7: Optical microscope image of the 64 kbit 1T1C FeRAM test chip.

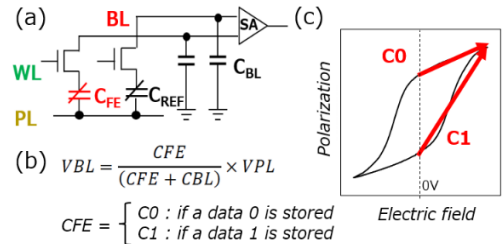


Fig. 8: Design of read out operation. (a) schematic of SA. 1T1C cell is connected to the bit line (BL), word line (WL), and plate line (PL). (b) A formula to determine the bitline voltage V_{BL} . (c) Definition of C_{FE} in (b).

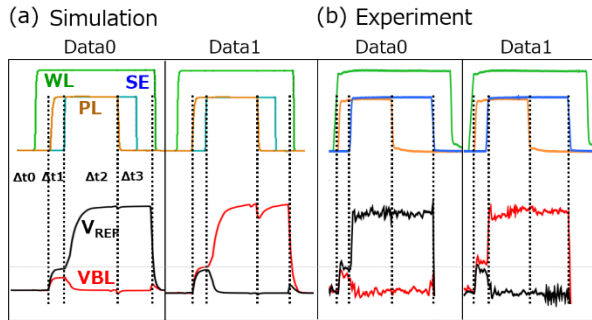


Fig. 9: (a) Timing diagram of a read operation as shown in Fig. 8. V_{BL} is precharged to 0 V during Δt_0 . V_{BL} is sensed and compared with reference voltage V_{REF} during Δt_1 . If the V_{BL} is larger than V_{ref} , it is activated to V_{DD} by sense enable pulse (SE) during Δt_2 turning the state of C_{FE} from data1 to data0. The data is written back to data1 during Δt_3 . (b) Wave forms of the read operation for data0 and data1 obtained by experiment for the test chip.

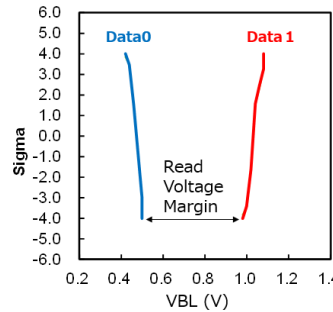


Fig. 10: V_{BL} distribution of the 1T1C FeRAM for 64kbit at 2.5 V with 100 ns operation at 1.0 μm^2 capacitor size of the test chip.

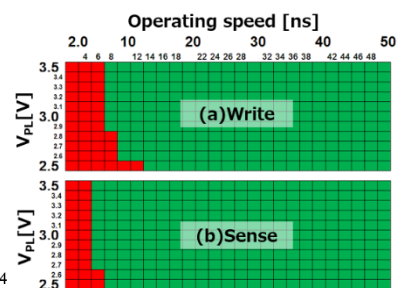


Fig. 11: Shmoo plot of (a)write and (b)sense time ranging from 1 ns to 50 ns and operation voltage ranging from 2.5 V to 3.5 V on the test chip. (Green: Pass, Red: Fail)

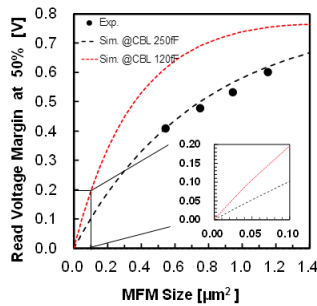


Fig. 12: Read margin of the test chip 2.5 V 100 ns operation as a function of capacitor size. Values of C_{BL} were determined as 250 fF for the 130nm technology and 120 fF for 40nm technology [6].

Table.1: Device structure comparison

Device	Volatile		Nonvolatile		
	SRAM	DRAM	FeFET [2]	FeRAM [3]	FeRAM This Work
Structure	6T	1T1C	1T	1T1C	1T1C
Write Voltage	~1.0 V	~1.0 V	4.2 V	4.0 V	2.5 V
Write Latency	<1ns	<20 ns	20 ns	100 ns	14 ns
Retention	Volatiles	Volatiles	250C, >7days	85C, >1000 min	85C, >100 min
Endurance (cycles)	>1E15	>1E15	1E5	>1E11	>1E11

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